

## HOW THE GENE-FAMILY COUPLING MODEL WORKS

### A Potts/Ising model of non-independent gene families in ZOMBI2

By default ZOMBI2 evolves every gene family independently, so the phylogenetic profile it emits—the *families*  $\times$  *species* presence/copy-number matrix—has rows that correlate *only* through the shared species tree. Real genomes correlate through *function*: families in one pathway or complex tend to be present or absent *together*, the signal that phylogenetic-profiling and inverse-Potts methods exploit. The **zombi2.coupling** model is the forward, *generative* counterpart: you prescribe pairwise couplings, and it produces profiles that carry that co-occurrence structure as a known ground truth for benchmarking those inference methods.

## 1 The model

Fix a panel of  $N$  gene families. Inside one genome, their presence/absence is an Ising vector  $\sigma \in \{0, 1\}^N$  ( $\sigma_i = 1$  means family  $i$  is present). Two ingredients set how a family behaves: a *field*  $h_i$  (its intrinsic tendency to be retained) and, for each pair, a *coupling*  $J_{ij}$  (how much  $i$  and  $j$  “like” to co-occur). Together they give the *local field* that family  $i$  feels from everyone else:

#### The two rate rules

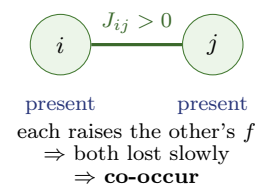
$$f_i(\sigma) = h_i + \sum_j J_{ij} \sigma_j \quad \Rightarrow \quad \underbrace{\text{loss}_i = \ell_0 e^{-\beta f_i}}_{\text{coupled}}, \quad \underbrace{\text{gain}_i = \text{horizontal transfer}}_{\text{field-blind}}$$

- **Loss carries the coupling.** A *present* family is lost at rate  $\ell_0 e^{-\beta f_i}$ . A present partner with  $J_{ij} > 0$  raises  $f_i$  and so *lowers*  $i$ ’s loss (they protect one another);  $J_{ij} < 0$  raises loss (they exclude one another);  $J_{ij} = 0$  is independence.  $h_i$  is the solo bias ( $h_i \gg 0$  is a near-universal “hub” gene);  $\beta$  is a global coupling strength.
- **Gain is horizontal transfer.** A lost family returns only when a donor that still has it transfers a copy in—the ordinary, function-blind HGT event.

The coupling never appears in the gain step. Instead it acts through *differential retention*: transfer sprays families around blindly, and the coupled loss then *keeps* a newly acquired family where its partners are present (high  $f_i$ , slow loss) and *purges* it where they are absent (low  $f_i$ , fast loss). That selective retention of horizontally-acquired genes is what writes  $J$  into the final profiles.

## 2 Why co-occurrence emerges

Take two families with a strong positive coupling. While both are present, each raises the other’s field, so both are lost slowly—they persist *together*. The moment one is lost, the other loses its protection, its loss rate jumps, and it tends to follow: losses *cascade*, giving all-or-nothing pathways. Across many lineages some keep the pair and some lose it, so the two rows of the profile line up—positive correlation. Flip the sign of  $J$  and the same logic runs backwards: each family’s presence *raises* the other’s loss, so they end up in different genomes (avoidance, negative correlation). With  $J = 0$  the rows are independent given the tree.



### 3 How it plugs into ZOMBI2

ZOMBI2’s forward simulator already hands each rate model the *whole* genome (through its `event_weights` method). The coupling is therefore a single new rate model, `PottsRates`, and **nothing else changes**—not the simulator, the genome, the event types, nor the output:

1. read the presence vector  $\sigma$  straight off the genome;
2. for every *present* family emit a loss weight  $\ell_0 e^{-\beta f_i}$ ; add one field-blind transfer channel for the gain side;
3. the simulator recomputes the branch’s weights after *every* event, so the field  $f_i$  is always current—i.e. genuine state-dependent (Glauber-style) dynamics along the tree, for free.

Because a custom rate model breaks per-family independence it runs on the pure-Python engine (the Rust fast path could not apply anyway); the cost is  $O(N + \text{nnz}(J))$  per event.

#### One caveat worth stating plainly

Because the gain channel is *field-blind*, the process has **no exact Boltzmann stationary distribution**—this is an *approximate* Potts generator by design (a lost gene returns only from a donor that has it, which is the honest biology). Consequently, couplings recovered from the profiles track the injected  $J$  in **sign and rank, not as an exact multiple**. Separately, on an ordinary birth–death tree even *uncoupled* families co-occur through shared ancestry (clade-restricted loss), so raw correlation is confounded by phylogeny—exactly why real inference corrects for the tree, and why the validation below uses a near-star tree to isolate the coupling.

### 4 Using it, and how we check it

```
from zombi2 import simulate_species_tree, BirthDeath, pathway_blocks, simulate_coupled

tree = simulate_species_tree(BirthDeath(1.0, 0.2), n_tips=100, age=6.0, seed=1)
spec = pathway_blocks([3, 3], within=3.0, between=-1.0, # two co-occurring pathways,
                      h=2.0, base_loss=1.0, transfer=0.2) # mutually exclusive
res = simulate_coupled(tree, spec, seed=1)
res.profiles # N x species matrix, ground-truth couplings baked in
```

Couplings can also be supplied as a dense matrix or a sparse edge list (`CouplingSpec.from_dense / .from_edges`). The test suite validates two things: (i) the loss rate equals  $\ell_0 e^{-\beta f_i}$  exactly—a positive partner lowers a family’s loss, a negative partner raises it; and (ii) *inject*  $\rightarrow$  *recover*—simulate under a known  $J$  and confirm the profiles show the prescribed structure ( $+J \Rightarrow$  co-occurrence,  $-J \Rightarrow$  avoidance,  $J = 0 \Rightarrow$  none), measured on a near-star tree so the signal is the coupling, not the phylogeny.